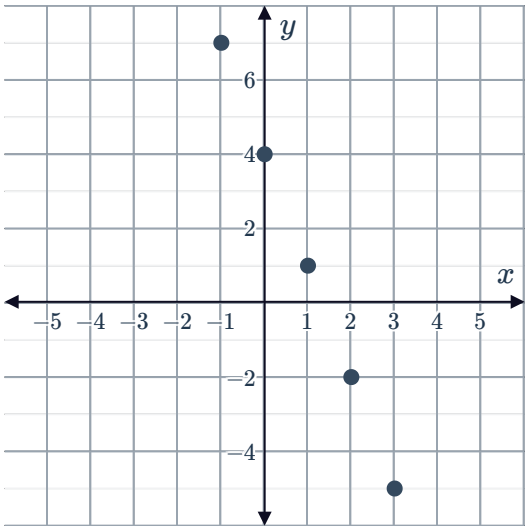
Appropriate models for data

1

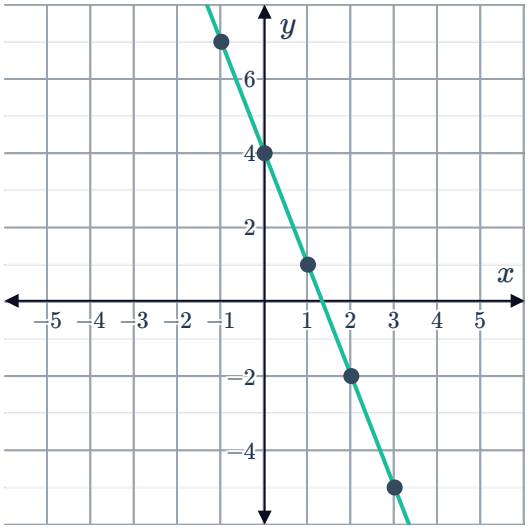
a

i



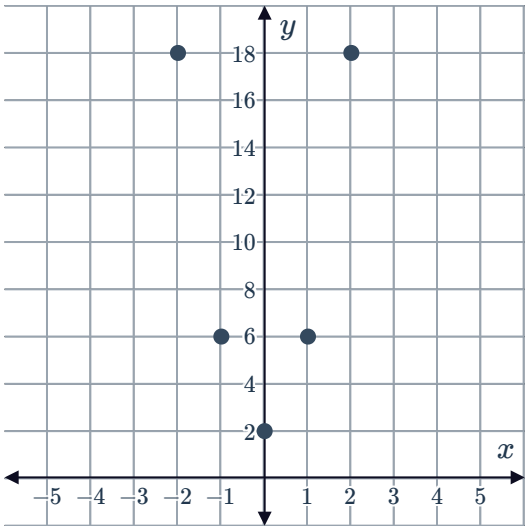
ii Linear

iii



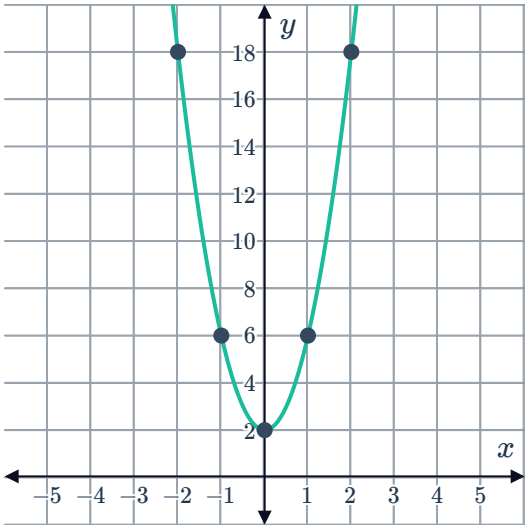
b

i



ii Quadratic

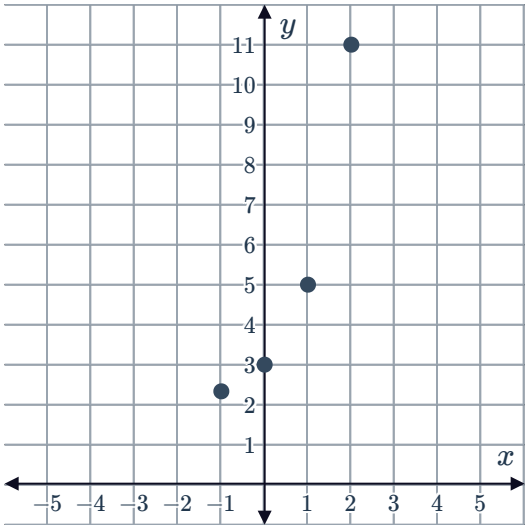
iii



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

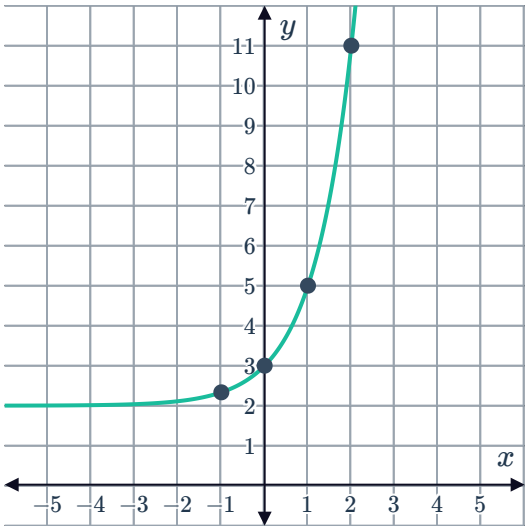
c

i



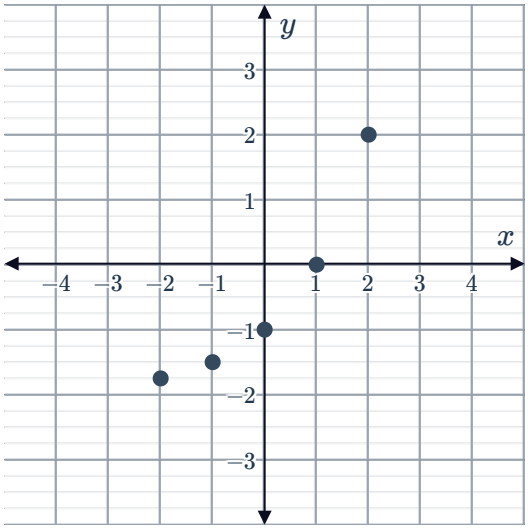
ii Exponential

iii



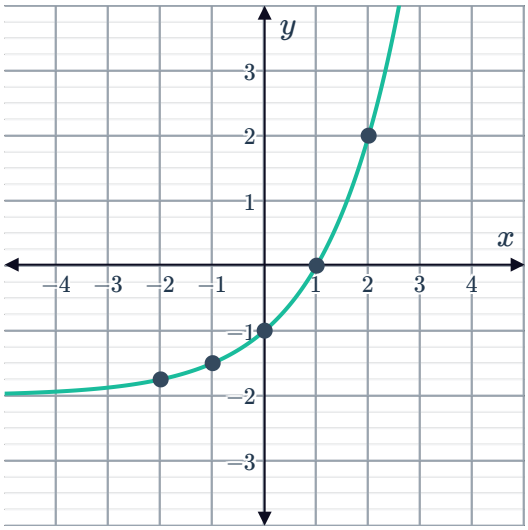
d

i



ii Exponential

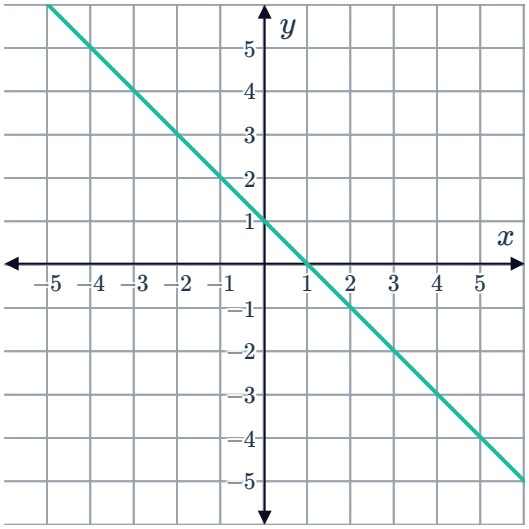
iii



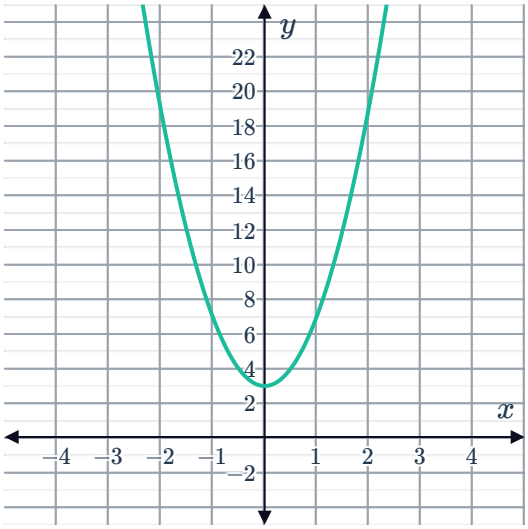
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

2

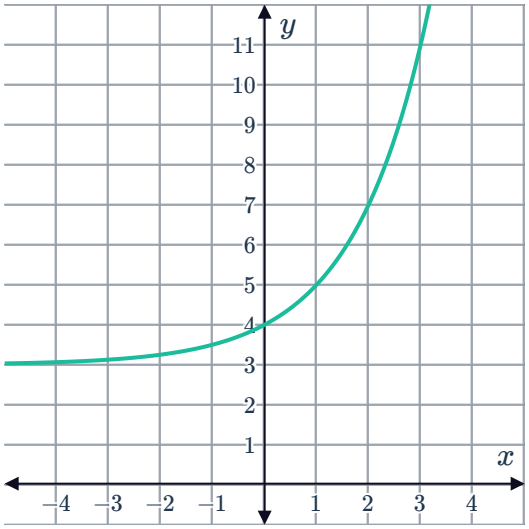
a



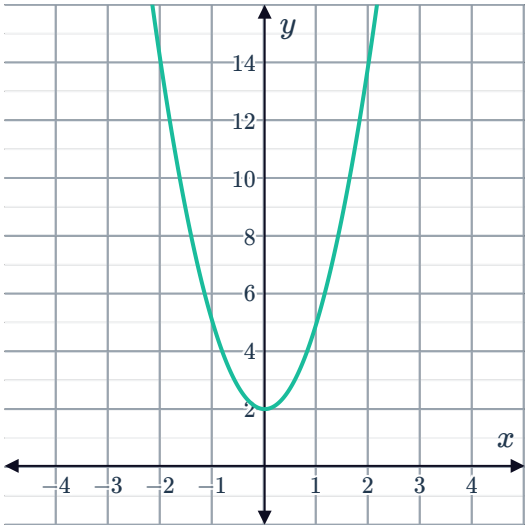
b



c



d



3

a A linear model would be the best way to model the data because the points are close to being along a straight line.

b A linear model would not be the best way to model the data because the points are close to being along a curve rather than along a straight line.

4

a Linear, $f(x) = mx + b$

b Quadratic, $f(x) = ax^2 + bx + c$, $a > 0$

c Quadratic, $f(x) = ax^2 + bx + c$, $a < 0$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Equations of models

5

a $y = a(x-h)^2 + k$

b $y = 2(x+3)^2 - 3$

6

a $y = a(x-p)(x-q)$

b $y = -(x-3)(x+1)$

7

a Cubic function

b **i** No **ii** Yes **iii** No **iv** Yes

c $2(x-1)(x-3)^2$

8

a Cubic

b **i** No **ii** Yes **iii** No **iv** No

c $y = -(x-1)(x-3)(x-4)$

9

a $y = a(x-h)^3 + k$

b $y = 2(x-1)^3 - 3$

10

a Rational

b **i** No **ii** No **iii** No **iv** Yes

c $y = \frac{2}{x} + 5$

11

a Rational

b **i** Yes **ii** No **iii** No **iv** No

c $y = -\frac{1}{x-4} + 3$

12

a $a = 5$ **b** $b = 2$ **c** $y = 5 \times 2^x$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13

a

i $c = 3$ **ii** $a = 3$ **iii** $b = 2$

b $y = 3 \times 2^{-x} + 3$

14

a $y = 2\log_{10} x$ **b** $y = \log_{10}(x - 2)$ **c** $y = \log_{10} 2x$

15

a

i Quadratic **ii** $y = \frac{x^2}{3} + 6x + 31$

b

i Rational **ii** $y = \frac{16}{x}$

Applications

16 Yes, for every week that passes, Rochelle's savings will increase by a constant amount.

17 Yes, the change in the volume of water in the bucket is constant over time.

18 No, for each unit of time, the change in height is not constant.

19 No, with every minute the change in temperature of the soup becomes smaller and smaller.

20

a

i No **ii** Yes **iii** No **iv** No

b 2

c 1 m

21

a 0

b No, there will ultimately be no Potassium-40 left in the sample.

22

a 0

b No, there is a minimum build time that will not decrease with ever more bricklayers.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23

a $24 \leq x \leq 30$

b $40 \leq x \leq 46$

c No - Jack's model is only accurate in the region $24 \leq x \leq 30$ and Georgia's model is only accurate in the region $40 \leq x \leq 46$, but there is a box used by the supermarket that has $x = 35$ which is in neither region.

24

a

i No **ii** No **iii** Yes **iv** No

b No. The data points outside the region $3 < x < 7$ begin to diverge from the linear model.

25

a

i No **ii** No

iii No **iv** Yes

b The tree will stop growing and remain at a height of $700\ \text{cm}$.

26

a 5 seconds

b The model fits the data before the truck reaches the town.

c Yes

27 B

28 C

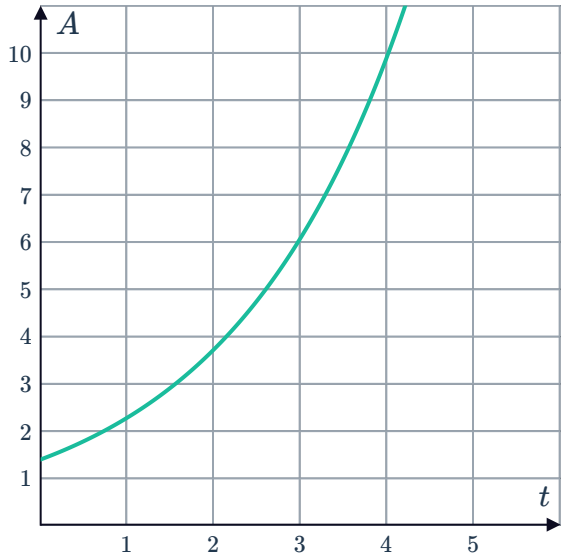
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

29

a

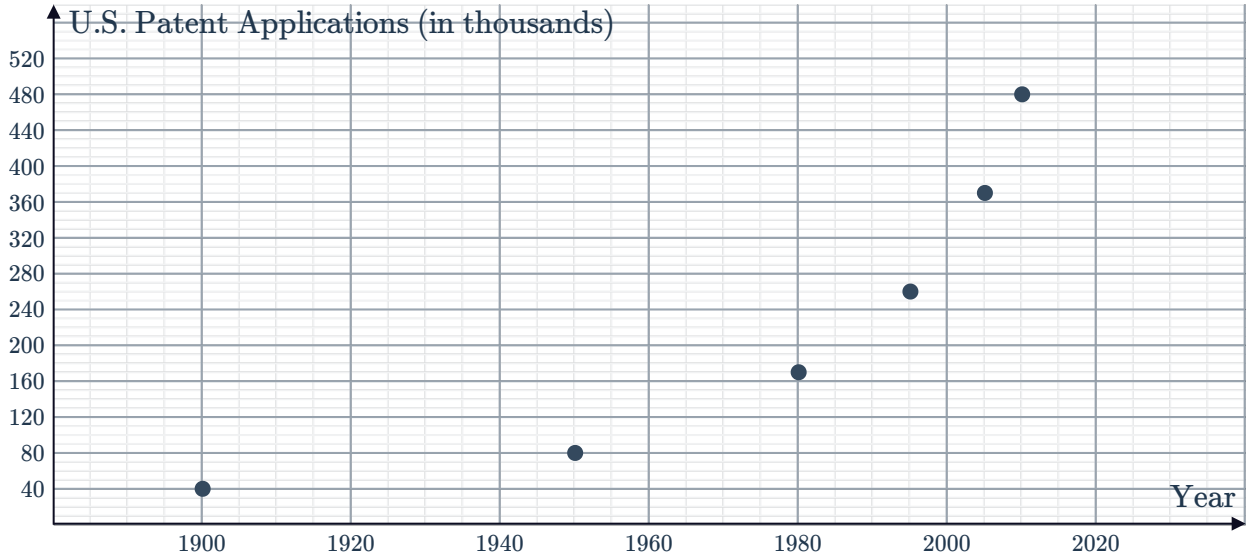
A	0	1	2	3
t	1.4	2.3	3.7	6.1

b



30

a

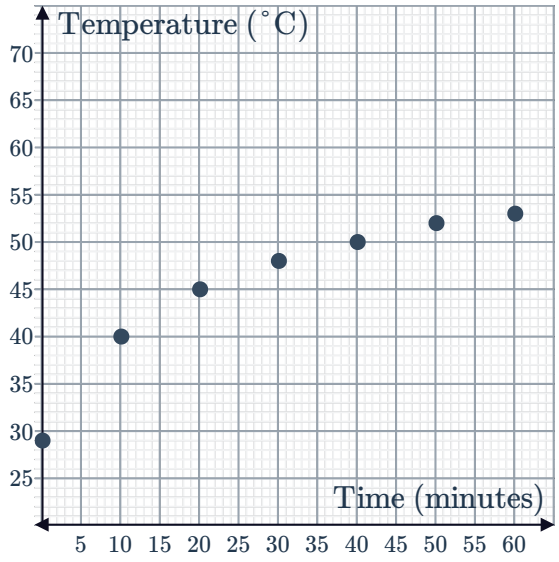


b Yes, an exponential model is appropriate because when drawn on a graph it has an exponential shape which, which is increasing more rapidly over time.
Also, if we calculate an exponential regression on this data set, we get $R^2 = 0.9627$ which is very close to 1, so there is a strong correlation to the exponential model suggesting an exponential association.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31

a



b A logarithmic function

© ClipSAT: By Mr. Mohamed Abdallah +966597688647